MATH329 · OPERATIONS RESEARCH

Quiz 3-4 Solutions

Big-M method · 30 minutes · term 471

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01 What is Big-M and why is it needed?

Simplex starts from the slack variables because they give an identity matrix. A `≥` constraint gets a surplus with a minus sign, which cannot be a starting basic variable. The fix is to add a temporary **Artificial variable**, written `R`, to that constraint.

To make sure `R` leaves the solution and becomes zero, we penalize it in the objective with a very large coefficient `M`:

Type	Penalty in the objective
Max	`− M R`
Min	`+ M R`

This explanation is mine, following the standard method in Taha's textbook, not your instructor's slides. His notation may differ.

02 Solution part 1: standard form

The first constraint is $\leq$, so add a slack. The second is $\geq$, so subtract a surplus.

$$\begin{aligned} &\text{Maximize } Z = 2x_1 + 3x_2 - 5x_3 + 0s_1 + 0s_2 \\ &\text{Subject to} \quad 4x_1 + 3x_2 - x_3 + s_1 = 24 \\ &\qquad\qquad x_1 + x_2 + 2x_3 - s_2 = 18 \\ &\qquad\qquad x_1, x_2, x_3, s_1, s_2 \geq 0 \end{aligned}$$

03 Solution part 2: Big-M

1 Add R to the second constraint and penalize it

$$\begin{aligned} &\text{Maximize } Z = 2x_1 + 3x_2 - 5x_3 + 0s_1 + 0s_2 - M\,R \\ &\text{Subject to} \quad 4x_1 + 3x_2 - x_3 + s_1 = 24 \\ &\qquad\qquad x_1 + x_2 + 2x_3 - s_2 + R = 18 \\ &\qquad\qquad x_1, x_2, x_3, s_1, s_2, R \geq 0 \end{aligned}$$

2 Write the tableau and fix the `Z` row

The `Z` row comes from `z − 2x₁ − 3x₂ + 5x₃ + M R = 0`:

Basic	x_1	x_2	x_3	s_1	s_2	R	Sol.
Z	−2	−3	5	0	0	M	0
s_1	4	3	−1	1	0	0	24
R	1	1	2	0	−1	1	18

The problem: `R` is basic but its coefficient in the `Z` row is `M`, and it must be zero. Fix it with:

```
New Z row = Z row − M × (R row)
```

3 Run simplex as usual

Basic	x_1	x_2	x_3	s_1	s_2	R	Sol.
Z	−M − 2	−M − 3	5 − 2M	0	M	0	−18M
s_1	4	3	−1	1	0	0	24
R	1	1	**2**	0	−1	1	18
Z	−9/2	−11/2	0	0	5/2	M − 5/2	−45
s_1	9/2	**7/2**	0	1	−1/2	1/2	33
x_3	1/2	1/2	1	0	−1/2	1/2	9
Z	18/7	0	0	11/7	12/7	M − 12/7	48/7
x_2	9/7	1	0	2/7	−1/7	1/7	66/7
x_3	−1/7	0	1	−1/7	−3/7	3/7	30/7

`M` is a very large number, so any coefficient containing `−M` counts as a large negative.

Iteration 1: the most negative entry is `5 − 2M` under `x3`. The `s1` row has coefficient `−1`, so it is skipped, and the `R` row has ratio `18/2 = 9`. So `R` leaves and the pivot is `2`.

Iteration 2: the most negative entry is `−11/2` under `x2`. The ratios are `33/(7/2) = 9.43` and `9/(1/2) = 18`. So `s1` leaves and the pivot is `7/2`.

All entries in the Z row are ≥ 0, and R = 0, so the solution is optimal.

Optimum Z = 48/7 = 6.86 at x_1 = 0, x_2 = 66/7 = 9.43, x_3 = 30/7 = 4.29

Check

```
Z = 3(66/7) − 5(30/7) = 198/7 − 150/7 = 48/7
```

```
4(0) + 3(66/7) − 30/7 = 168/7 = 24        ≤ 24   true
```

```
0 + 66/7 + 2(30/7) = 126/7 = 18           ≥ 18   true
```

The quiz came with no answer key. This solution is my own, and I checked it by computer.